Haodong Lei

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Education

Southeast University — Software Engineering — Sep 2024 - Jun 2027

• Member of Pattern Learning and Mining (PALM) Lab; Research Focus: Agent Memory, LLM Acceleration

• **Scores: 85.85; TOEFL iBT (Updated):** Overall 4.5 (R5.0, L4.5, S4.0, W5.0), Test Date: 2026-10-14.

Hainan University — Computer Science and Technology — Sep 2020 - Jun 2024

• **GPA:** 3.88/4.0, **Scores: 90.96;** Overall Ranking: 5/123 (4.06%)

Research Experience

- AI Agent — Oct.2025-Present
- AI Memory — Oct.2025-Present
- LLM Reasoning — Oct.2025-Present
- LLM Speculative Sampling — Oct.2024-Oct.2025

Publications (*denotes equal contribution, †denotes corresponding authors)

- **H Lei**, Y Chen, P Cai, B Shi, D Wang*†, H Wang*†, et al."TransMem: Transforming Hidden States into Memory for Large Language Models" — First Author, Under Review
- **H Lei**, D Wang*†, H Wang*†, et al."MemCoT: Test-Time Scaling through Memory-Driven Chain-of-Thought" — First Author, Under Review
- **H Lei**, H Wang†, et al. "Parallel-Path Relaxed Speculative Jacobi Decoding for Accelerating Auto-Regressive Text-to-Image Generation" — First Author, Under Review
- **H Lei**, H Wang†, et al. "Fast Inference of Visual Autoregressive Model with Adjacency-Adaptive Dynamical Draft Trees" — First Author, Under Review
- X Tan*, W Weng*, **H Lei**, et al. "EasyTune: Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation." — Third Author, ICLR26

Internship Experience

Shanghai AI Lab — Shanghai AI Lab — Oct 2025 - Present — Shanghai

- **AI Agent Long-Context Memory:** Designed MemCoT, a multi-scale memory system for test-time inference. Addressed challenges in LoCoMo dataset with 300+ turns, 9K+ tokens cross-month dialogues. Proposed Memory-Thought self-evolution framework with multi-scale long-term memory perception, integrating fine-grained retrieval and coarse-grained observation. Introduced dual short-term memory (semantic and situational trajectories), enabling agents to dynamically decompose and prune complex queries based on historical failure traces. Achieved SOTA on LoCoMo, improving F1 from 42.25 to 58.03 with GPT-4o-mini compared to pure RAG baseline. (**MemCoT, AAAI 2027 Phase 2)**
- **AI Learnable Latent Memory:** We introduce TransMem, a parametric memory module designed to enhance long-context reasoning at inference time by transforming historical hidden states into reusable memory representations. TransMem keeps the backbone LLM frozen and is invoked only when long-context reasoning is required. It is trained through self-supervision to identify and exploit relevant evidence in lengthy contexts, thereby improving

the LLM's utilization of historical information. TransMem delivers substantial gains on long-context reasoning benchmarks, including LoCoMo, HotpotQA, and MemoryAgentBench, improving over the corresponding backbones by 11.58–29.25 F1 points on LoCoMo and 10.20–13.03 F1 points on HotpotQA, while increasing the average accuracy on MemoryAgentBench from 29.54% to 40.00%. (TransMem, Under Review)

Research Experience

1、Text-to-Image Acceleration Southeast University & SMU Jul 2025 - Mar 2026

- **Extended speculative sampling draft tree:** Introduced multi-sequence draft tree structure for slow autoregressive text-to-image generation, expanding traditional SJD chain dependencies to hierarchical tree search, significantly improving Draft Token acceptance rate. Paper is under review.

  Relaxed sampling for image generation: Proposed Cross-Relaxed Strategy leveraging semantic similarity between sequences to optimize relaxed sampling distribution. Achieved approximately 4.14x to 4.18x speedup on Parti-Prompts and MSCOCO2017 datasets. Paper is under review. (PathSpec, Under Review)
- **Similarity-based acceleration for image generation:** Proposed tree-based Speculative Sampling addressing similarity issues in adjacent draft trees. Dynamically adjusted draft tree shape based on different pixel block depths and widths. Designed Adaptive Draft Tree method for Visual AR model Speculative Sampling, enabling Anole model to achieve acceptance length of 3.4 on MS-COCO with 2.21x inference speedup compared to baseline. Paper is under review. (ADT-Tree, Under Review)

3、Text-to-Motion Generation Southeast University & SMU Dec 2024 - Nov 2025

- **Diffusion model post-training:** Proposed EasyTune, a fine-tuning framework for diffusion models achieving (1) dense efficient optimization, (2) memory-efficient training, and (3) fine-grained alignment by decoupling recursive dependencies. Paper has been accepted to ICLR26

Academic Service

Conference Reviewer: ACMMM2025, ICLR 2026, AAAI 2027, ICLR 2027

Language

- **TOEFL iBT (Updated): Overall 4.5 (R5.0, L4.5, S4.0, W5.0), Test Date: 2026-10-14.**

Skills

- Large model lightweight algorithms: Speculative sampling
- AI Agent Memory construction: RAG, knowledge graphs, Multi-Agent systems

Awards

First Prize, 2023 Cross-Strait Computer Innovation Competition, National Level 2023.06

Third Prize, 25th China Robot and AI Competition - AI Creative Track, National Level 2023.06